

Chapter-1 (Permutation and Combination)

Basic Principle of Counting:

Multiplication Principle:

If an operation can be performed in m different ways, following which a second operation can be performed in n different ways, the whole operation can be performed in $m \times n$ different ways.

Addition Principle

If an operation can be performed in m different ways and another operation can be performed in n different ways, exactly one of the operations can be performed in $m + n$ ways.

Factorial:

The falling product of natural number is called factorial. It is denoted by $n!$ and read as factorial n .

So, $1! = 1$; $2! = 2 \times 1$; $3! = 3 \times 2 \times 1$

$n! = n \times (n-1) \times \dots \times 3 \times 2 \times 1$

Thus, $n! = n \times (n-1)!$

Note: $0! = 1$

Permutations:

Permutation is the arrangement of objects in a definite order taking some or all at a time.

-The number of permutation of n objects taken r ($r \leq n$) at a time is denoted by nPr or $P(n, r)$.

Example: The permutation of the three letters a, b, c taken two at a time are ab, ba, bc, ca, bc, cb.

Various types of permutations:

Case I: Permutation without repetition:

Theorem: The total number of permutations of a set of n distinct objects taken r at a time is given by

$$P(n, r) = n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!}, \quad (0 \leq r \leq n)$$

Note:

- i. $P(n, n) = n!$
- ii. $P(n, 0) = 1$
- iii. $P(n, 1) = n$
- iv. $\frac{P(n, r)}{P(n, r-1)} = n - r + 1$

Case II: Permutation of objects not all different:

Theorem: The total number of permutations of a set of n distinct objects taken all at a time if there are p objects of 1st kind q objects of 2nd kind and r objects of 3rd kind is $\frac{n!}{p! q! r!}$.

Case III: Circular Permutation:

Theorem: The total number of permutations of a set of n distinct objects arranged in a circle is $(n-1)!$.

Note: In case of necklace of beads, beads into a bracelet, the required number of arrangements in a circle is $\frac{1}{2}(n-1)!$

Case IV: Permutation of repeated things:

Permutations of a set of n objects taken r at a time with repetition is n^r .

Combinations:

-Combination is the selection of objects taken some or all at a time.

-The number of combinations of n distinct objects taken r at a time is denoted by nCr or $C(n, r)$.

Example: The combination of the three letters a, b, c taken two at a time are ab, bc, ca.

Theorem: The total number of combination of the set of n objects taken r at a time is given by

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Some Properties of Combinations:

1. Complementary relation:

$$C(n, r) = C(n, n-r)$$

2. Corollary : If $C(n, r_1) = C(n, r_2)$ then either $r_1 = r_2$ or, $r_1 + r_2 = n$

3. Pascal's rule:

$$C(n, r-1) + C(n, r) = C(n+1, r)$$

Note:

1. $\frac{C(n, r)}{C(n, r-1)} = \frac{n-r+1}{r}$
2. $C(n, 0) + C(n, 1) + \dots + C(n, n) = 2^n$
3. The combinations of n objects taking r at a time if $(q < r)$ particular objects are always included is $C(n-q, r-q)$
4. The combinations of n objects taking r at a time if $(q < r)$ particular objects are always excluded is $C(n-q, r)$

Some Key points for MCO's:

1. The relation between permutation and combination is $P(n, r) = r! \cdot C(n, r)$
2. The number of ways in which n distinct objects can be put into m distinct places is m^n
3. If a polygon has n distinct sides, then the number of diagonals is : $\frac{n(n-3)}{2} = C(n, 2) - n$

- The number of lines that can be drawn by using n points in a plane of which p points are collinear is: $C(n, 2) - C(p, 2) + 1$
- In n peoples shakes hand with other, then the total number of hand shaken is $C(n, 2)$.
- Number of triangles that can be formed from n points in which m points are collinear is $C(n, 3) - C(m, 3)$

Chapter-2 (Binomial Theorem)

Binomial Theorem:

For any positive integer n ,

$$(a + x)^n = C(n, 0)a^n + C(n, 1)a^{n-1}x + \dots + C(n, r)a^{n-r}x^r + \dots + C(n, n)x^n$$

$$\text{i.e. } (a + x)^n = \sum_{r=0}^n C(n, r) a^{n-r} x^r$$

Note:

- The expression $(a + x)^n$ contains $(n + 1)$ terms
- Sum of binomial exponent in each term is n

General term:

The general term, $(r + 1)^{th}$ term of the expansion of $(a + x)^n$ is: $t_{r+1} = C(n, r)a^{n-r}x^r$

Middle terms:

The total number of terms after the expansion of $(a + x)^n$ is $(n + 1)$. Thus, if n is even $n + 1$ is odd and the expression contain single middle term. But, if n is odd then $n + 1$ is even and hence the expansion contains two middle terms.

Case I: If n is even, the middle term in $(a + x)^n$ is obtained by putting $r = \frac{n}{2}$,

Case I: If n is odd, there are two middle terms which are obtained by putting $r = \frac{n-1}{2}$ and $\frac{n+1}{2}$

Note:

The greatest value of $n(C, r)$ is the coefficient of middle term.

Binomial Coefficients:

The coefficients $C(n, 0), C(n, 1), C(n, 2), \dots, C(n, n)$ in the expansion of $(a + x)^n$ are known as binomial coefficients.

$$C_0 = C(n, 0) = 1; C_1 = C(n, 1) = \frac{n}{1};$$

$$C_2 = C(n, 2) = \frac{n(n-1)}{2!}; C_3 = C(n, 3) = \frac{n(n-1)(n-2)}{3!}$$

Notes:

→ Sum of binomial coefficients in the expansion of $(1 + x)^n$ is 2^n .

$$\text{i.e. } C_0 + C_1 + C_2 + \dots + C_n = 2^n$$

→ Sum of all even binomial coefficients = sum of all odd binomial coefficients = 2^{n-1}

$$C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots = 2^{n-1}$$

→ The total number of terms in the trinomial expansion $(ax + by + cz)^n$ is $\frac{(n+1)(n+2)}{2}$

Notes:

$$\text{i. } 1 + n + \frac{n(n-1)}{2!} + \dots + 1 = (1 + 1)^n$$

$$\text{ii. } 1 - n + \frac{n(n-1)}{2!} - \dots + (-1)^n 1 = (1 - 1)^n$$

$$\text{iii. } 1 + (n-1) + \dots + 1 = (1 + 1)^{n-1}$$

$$\text{iv. } 1 - (n-1) + \dots + (-1)^{n-1} \cdot 1 = (1 - 1)^{n-1}$$

Binomial Theorem for any rational numbers

If n is negative integer or a rational fraction, then

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Thus the expansion is infinite.

Which is valid only if $|x| < 1$

Exponential Series:

The exponential series expansion is defined as e^x and is defined as

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \text{ to } \infty = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Notes:

$$1. e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \text{ to } \infty$$

$$2. e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \infty$$

$$3. e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots +$$

$$4. \frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \text{ to } \infty = \cos hx$$

$$5. \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \text{ to } \infty = \sin hx$$

$$6. \frac{e^1 + e^{-1}}{2} = 1 + \frac{1}{2!} + \frac{1}{4!} + \dots \text{ to } \infty = \sum_{n=0}^{\infty} \frac{1}{(2n)!}$$

$$7. \frac{e^1 - e^{-1}}{2} = 1 + \frac{1}{3!} + \frac{1}{5!} + \dots \text{ to } \infty = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!}$$

$$8. \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e = \lim_{n \rightarrow 0} (1 + n)^{\frac{1}{n}}$$

$$9. e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{nx}$$

$$10. a^x = 1 + \frac{x \log a}{1!} + \frac{(x \log a)^2}{2!} + \frac{(x \log a)^3}{3!} + \dots$$

Sum of the series whose n^{th} term are known:

1. $\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e$; $\sum_{n=1}^{\infty} \frac{a^n}{(n-1)!} = e^a$
2. $\sum_{n=1}^{\infty} \frac{1}{n!} = e - 1$; $\sum_{n=1}^{\infty} \frac{a^n}{n!} = e^a - 1$
3. $\sum_{n=1}^{\infty} \frac{1}{(n+1)!} = e - 2$; $\sum_{n=1}^{\infty} \frac{a^n}{(n+1)!} = e^a - 2$
4. $\sum_{n=1}^{\infty} \frac{n^2}{(n)!} = 2e$; $\sum_{n=1}^{\infty} \frac{n^3}{(n)!} = 5e$

Logarithmic Series:

1. $\ln(1+x) = \log_e(1+x)$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \text{ to } \infty$$

$$-1 < x \leq 1$$
2. $\ln(1-x) = \log_e(1-x)$

$$= -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots \text{ to } \infty$$

$$-1 \leq x < 1$$
3. $\ln\left(\frac{1+x}{1-x}\right) = 2\left(1 + \frac{x^3}{3} + \frac{x^5}{5} + \dots \text{ to } \infty\right)$; $|x| < 1$
4. $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots \text{ to } \infty$
5. $\ln 2 = \frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \dots \text{ to } \infty$

Chapter:3

Sequence and Series

Review:

Arithmetic Progression(A.P)

→ If a is the first term and d be the common difference of an A.P then

→ General term / last term / n^{th} term

$$(t_n) = a + (n-1)d$$

→ Sum to first n terms $(S_n) = \frac{n}{2} [2a + (n-1)d]$

→ If b is the last term of an A.P. then

$$S_n = \frac{n}{2} [a + b]$$

Geometric Progression(G.P)

→ Let a be the first term and r be the common ratio of G.P then

→ General term / last term / n^{th} term $(t_n) = ar^{n-1}$

→ Sum to first n terms $(S_n) = \frac{a(r^n - 1)}{r - 1}$ if $r > 1$

$$\text{or } S_n = \frac{a(1 - r^n)}{1 - r} \text{ if } r < 1$$

→ If b is the last term of a G.P. then

$$S_n = \frac{(br - a)}{r - 1} \text{ if } r > 1 \text{ or } S_n = \frac{(a - br)}{1 - r}, \text{ if } r < 1$$

Arithmetico- Arithmetic Series (A-A Series)

The series obtained by multiplying the corresponding terms of two arithmetic series is called an A-A series.

Example: $1.3 + 2.4 + 3.5 + \dots$

Alternatively,

A series is said to be an A-A series if the series itself is not an arithmetic series but the difference of two consecutive terms form an arithmetic series.

Example: $1 + 3 + 6 + 10 + \dots$

Arithmetico-Geometric Series (A-G Series)

The series obtained by multiplying the terms of an arithmetic series with the corresponding terms of geometric series is called A-G series.

Example: $1.2 + 2.2^2 + 3.2^3 + \dots$

Sum to n terms of some special series:

1. Sum of first n natural numbers

$$1 + 2 + 3 + \dots + n = \sum n = \frac{n(n+1)}{2}$$
2. Sum of first n odd natural numbers

$$1 + 3 + 5 + \dots + (2n-1) = \sum (2n-1) = n^2$$
3. Sum of first n even natural numbers

$$2 + 4 + 6 + \dots + 2n = \sum (2n) = n(n+1)$$
4. Sum of squares of first n natural numbers

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$
5. Sum of cubes of first n natural numbers

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \sum n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Chapter-4

(Complex Number)

Complex Number:

→ A number of the form $a + ib$ where a and b both are real numbers, is called complex number,

Where, ' i ' is called the imaginary unit or iota.

$$[i = \sqrt{-1} \text{ i.e } [i = 0 + 1i = (0, 1)]]$$

→ It is generally denoted by Z . If $Z = a + ib$ then ' a ' is called real part and ' b ' is called imaginary part of the complex number Z .

→ The set of complex number is denoted by C and defined by $C = \{(a, b) : a, b \in R\}$

Notes:

- i) The set of real number is the subset of complex number.
- ii) The complex number don't satisfy the order property.
- iii) The complex conjugate or simply conjugate of $Z = a + ib$ is $\bar{Z} = a - ib$

Absolute value of complex number:

The absolute value of a complex number $Z = a + ib$ is denoted by $|Z|$ and defined by $|Z| = \sqrt{a^2 + b^2}$

Cube roots of unity:

The cube roots of unity are $1, w, w^2$. Where

$$z = \frac{-1 + i\sqrt{3}}{2} = w \text{ or } w^2 ; Z = \frac{-1 - i\sqrt{3}}{2} = w^2 \text{ or } w$$

Notes:

- i) Among 3 roots, 1 is the real cube root of unity but w and w^2 are the imaginary cube roots of unity.
- ii) The sum of 3 cube roots of unity is zero.
i.e. $1 + w + w^2 = 0$
So, i) $1 + w = -w^2$
ii) $1 + w^2 = -w$
iii) $w + w^2 = -1$
- iii) Product of imaginary cube roots of unity is 1.
 $w \cdot w^2 = 1$ i.e. $w^3 = 1$
- iv) Each imaginary cube root of unity is square of other.
i.e. $(w^2)^2 = w$
- v) The square roots of w are $\pm w^2$ and the square roots of w^2 are $\pm w$
- vi) Cube roots of unity in complex plane forms equilateral triangle
- vii) $|w^n| = 1$. So, $1 + |w| + |w^2| = 3$

Different Forms of Complex Numbers:

i) Cartesian form:

The Cartesian form of complex number is

$$Z = a + ib$$

ii) Polar form:

The polar form of the complex number $Z = a + ib$ is

$$Z = r (\cos\theta + i\sin\theta)$$

Where, $r = \sqrt{x^2 + y^2}$; $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

'r' is known as the modulus or magnitude of the complex number and θ is known as the argument or amplitude of Z .

iii) Exponential form (Euler's Formula)

The exponential (Euler) form of the complex number $Z = a + ib$ is $Z = re^{i\theta}$

$$\text{Where, } e^{i\theta} = (\cos\theta + i\sin\theta)$$

Notes:

- i) The principle value of argument of Z is given by $\arg(Z) = \tan^{-1}\left(\frac{y}{x}\right)$; $Z \neq 0$; $-\pi \leq \theta \leq \pi$
- ii) The principle value of argument of Z will be θ , $\pi - \theta$, $-(\pi - \theta)$ and $-\theta$ according as the point Z lies in the 1st, 2nd, 3rd and 4th quadrant respectively.
- iv) If $\arg(Z) = 0$ or π then Z is purely real
- v) If $\arg(Z) = \frac{\pi}{2}$ or $\frac{-\pi}{2}$ then Z is purely imaginary
- vi) Argument of a complex number 0 is not defined
- vii) The modulus of $e^{i\theta}$ is 1
i.e. $e^{i\theta} = |e^{i\theta}| = |\cos\theta + i\sin\theta| = 1$
- viii) For the complex numbers, $Z = a$; $Z = -a$; $Z = ib$; $Z = -ib$ the value of arguments are 0 ; π ; $\frac{\pi}{2}$; and $\frac{3\pi}{2}$ respectively.

Product and Quotient of Complex Number:

Let $Z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and

$Z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ then

- i) $Z_1 \cdot Z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$
- ii) $\frac{Z_1}{Z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]$

De Moivre's Theorem:

For any real number n (positive or negative or zero)

$$[r(\cos\theta + i\sin\theta)]^n = r^n (\cos n\theta + i\sin n\theta)$$

Nth roots of Complex number:

For any non-zero complex number

$Z = r(\cos\theta + i\sin\theta)$ and any positive integer n ,

the n distinct roots are given by

$$Z_k = \sqrt[n]{r} \left[\cos\left(\frac{\theta + k \cdot 360}{n}\right) + i\sin\left(\frac{\theta + k \cdot 360}{n}\right) \right]$$

where ; $k = 0, 1, 2, \dots, n - 1$

Chapter-5

(Matrix Based System of Linear Equations)

Linear Equation:

An equation of the form $ax + by = c$, where at least one of the constants a, b is non zero, is called a linear equation in two variables x and y .

System of Linear Equations:

A set of two linear equations

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

is called the system of linear equations in two variables x and y .

Note:

A system of linear equation may have

- Unique solution
- No solution
- Infinitely many solutions

→The system of two linear equations in two variables x and y

i.e
$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

has

- Unique solution : if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
- No solution if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
- Infinitely many solutions if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Notes: A system of linear equations is sometimes called simultaneous linear equations.

Definition:

A system of linear equations is said to be **consistent** if it has at least one solution. Otherwise, system is said to be **inconsistent**.

Definition

A system of linear equations is said to be **independent** if it has at most one solution. Otherwise, system is said to be **dependent**.

Consistent and Independent System

A system of linear equations is said to be consistent and independent if the system has exactly one solution.

Example: $x + y = 5; x - y = 3$

Inconsistent and Independent System

A system of linear equations is said to be inconsistent and independent if the system has no solution.

Example: $x + y = 1, x + y = 5$

Consistent and Dependent System

A system of linear equations is said to be consistent and dependent if there are infinitely many solutions of the system.

Example: $-x + 2y = 2; 3x - 6y = -6$

Cramer's Rule:

The rule of solving a system of linear equations by calculating the values of determinant formed by coefficients and constant is known as **Cramer's rule or the method of determinant**.

Solution of system of linear equations by Cramer's rule

→Consider the system of linear equations in two variables x and y

i.e
$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

Then by Cramer's rule,

$$x = \frac{D_x}{D} \text{ and } y = \frac{D_y}{D}$$

→Consider the system of linear equations in three variables x, y and z

i.e
$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

Then by Cramer's rule,

$$x = \frac{D_x}{D}, y = \frac{D_y}{D}; z = \frac{D_z}{D}$$

Notes:

i) If $D \neq 0$, then the system is **consistent and independent**

i.e. the system has a unique solution.

(ii) If $D = 0$ but either of D_x or D_y or D_z is not zero, then the system is **inconsistent and independent**.

i.e the system has no solution.

(iii) If $D = D_x = D_y = D_z = 0$, then the system is **consistent and dependent**.

i.e the system has infinite number of solutions.

Row Equivalent Matrix Method:

Elementary Row Operations: (ERO)

- i) Interchange of any two rows, i.e $R_i \leftrightarrow R_j$
- ii) Multiplication of any row by a non-zero number,

$$R_i \rightarrow kR_i$$

- iii) Multiple of a row added to any other row

$$R_i \rightarrow R_i + kR_j$$

Row Equivalent Matrix

A matrix 'A' is said to be row equivalent to a matrix B and we write $A \sim B$ if B can be obtained from A performing one or more of the elementary row operations.

Note: If $A \sim B$, then the system represented by A and the system represented by B have same solutions.

Inverse Matrix Method

The method of solving a system of linear equations by using the inverse of the coefficient matrix of the system is called inverse matrix method.

Thus the method is also known as ***Inversion method*** or simply a ***matrix method***.

Formula:

$$X = A^{-1}B ; \text{ where } A^{-1} = \frac{1}{|A|} \text{ Adjoint (A)}$$

Adjoint of A = Transpose of cofactor matrix.

Thank You